

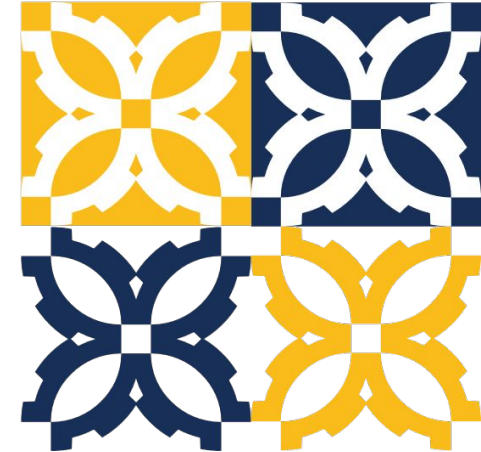


Pemrograman Perangkat Bergerak

ITS - Informatics Bachelor Study Program

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Android Kamera

Terdapat **dua cara**, menggunakan kamera dalam Android:

1. Menggunakan **aplikasi kamera** android di aplikasi kita
2. Menggunakan **Camera API** yang disediakan oleh android di aplikasi kita



Menggunakan aplikasi kamera android

Menggunakan MediaStore.ACTION_IMAGE_CAPTURE

Intent intent = new Intent(android.provider.MediaStore.ACTION_IMAGE_CAPTURE);

Beberapa data di MediaStore antara lain:

No	Jenis maksud dan deskripsi
1	ACTION_IMAGE_CAPTURE_SECURE mengembalikan gambar yang diambil dari kamera, saat perangkat ini diprotect
2	ACTION_VIDEO_CAPTURE panggilan aplikasi video yang ada di android untuk menangkap video
3	EXTRA_SCREEN_ORIENTATION mengatur orientasi layar untuk vertikal atau landscape
4	EXTRA_FULL_SCREEN mengontrol user interface dari ViewImage
5	INTENT_ACTION_VIDEO_CAMERA meluncurkan kamera dalam modus video
6	EXTRA_SIZE_LIMIT menentukan batas ukuran video atau pengambilan gambar ukuran

Pemanggilan Intent

Intent intent = new
Intent(android.provider.MediaStore.ACTION_IMAGE_
CAPTURE);

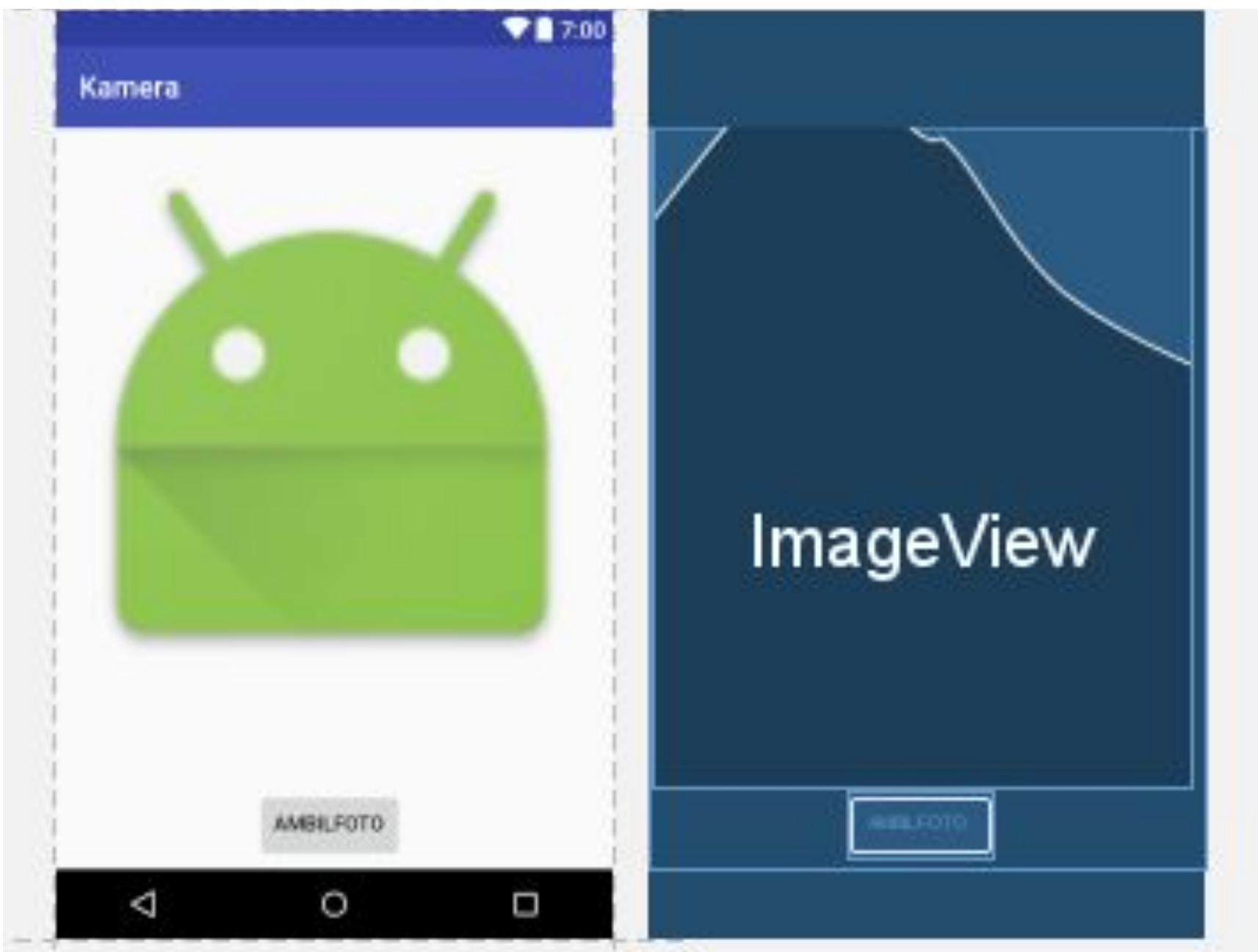
startActivityResult(intent,o)

This method has been defined in the **activity** class. We are calling it from main activity. There are methods defined in the activity class that does the same job , but used when you are not calling from the activity but from somewhere else. They are listed below

Beberapa pilihan menjalankan akitifitas antara lain

No	Activity function description
1	startActivityResult(Intent intent, int requestCode, Bundle options) It starts an activity , but can take extra bundle of options with it
2	startActivityFromChild(Activity child, Intent intent, int requestCode) It launch the activity when your activity is child of any other activity
3	startActivityFromChild(Activity child, Intent intent, int requestCode, Bundle options) It work same as above , but it can take extra values in the shape of bundle with it
4	startActivityFromFragment(Fragment fragment, Intent intent, int requestCode) It launches activity from the fragment you are currently inside
5	startActivityFromFragment(Fragment fragment, Intent intent, int requestCode, Bundle options) It not only launches the activity from the fragment , but can take extra values with it

Pembuatan Aplikasi Sederhana



```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
```

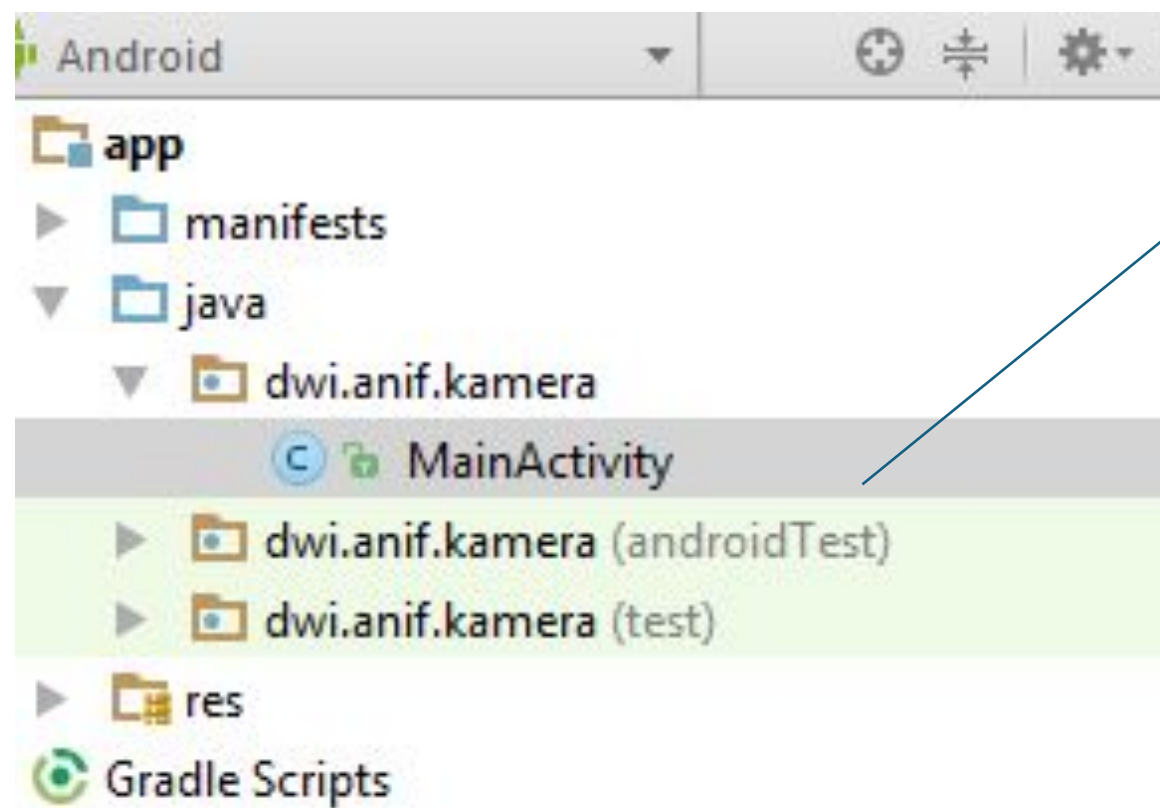
```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context="dwi.anif.kamera.MainActivity">
```

```
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="AmbilFoto"
    tools:layout_editor_absoluteX="138dp"
    tools:layout_editor_absoluteY="456dp"
    tools:ignore="MissingConstraints" />
```

```
<ImageView
    android:id="@+id/imageView"
    android:layout_width="372dp"
    android:layout_height="514dp"
    app:srcCompat="@mipmap/ic_launcher"
    tools:layout_editor_absoluteY="-58dp"
    android:layout_marginRight="8dp"
    app:layout_constraintRight_toRightOf="parent"
    tools:ignore="MissingConstraints" />
```

```
</android.support.constraint.ConstraintLayout>
```

Pembuatan Aplikasi Sederhana



```
package dwi.anif.kamera;

import android.content.Intent;
import android.provider.MediaStore;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;

public class MainActivity extends AppCompatActivity {
    Button b1;
    ImageView iv;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1 = (Button) findViewById(R.id.button);
        iv = (ImageView) findViewById(R.id.imageView);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new
                Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                startActivityForResult(it, 0);
            }
        });
    }
}
```

Jalankan program, maka akan muncul button, jika ditekan, maka fungsi kamera akan aktif dan jika dilakukan pengambilan gambar, maka hasil akan bias di lihat pada file di gallery

Menyimpan Foto ke Tempat Lain

Dengan memakai program dibawah ini, sebenarnya hasil foto sudah terdapat di file gallery

```
Intent it = new
Intent(MediaStore.ACTION_IMAGE
_CAPTURE);
```

```
startActivityResult(it, 0);
```

Jika di inginkan hasil tersebut berada di folder atau file lain, maka diperlukan tempat yang harus di tentukan. Contoh program ini ada pada directory DICM

```
public class MainActivity extends AppCompatActivity {
    Button b1;
    ImageView iv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1 = (Button) findViewById(R.id.button);
        iv = (ImageView) findViewById(R.id.imageView);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                File imagesFolder =
Environment.getExternalStoragePublicDirectory
(Environment.DIRECTORY_DCIM);
                Date d = new Date();
                CharSequence s =
DateFormat.format("MM-dd-yy hh-mm-ss", d.getTime());
                File image = new File(imagesFolder, s.toString() + ".jpg");
                Uri uriSavedImage = Uri.fromFile(image);
                it.putExtra(MediaStore.EXTRA_OUTPUT, uriSavedImage);
                startActivityResult(it, 0);
            }
        });
    }
}
```

Menyimpan dg memakai
putExtra

Menyimpan Foto ke Tempat Lain (2)

```
public class MainActivity extends AppCompatActivity {
    Button b1;
    ImageView iv;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        b1 = (Button) findViewById(R.id.button);
        iv = (ImageView) findViewById(R.id.imageView);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                File imagesFolder = new File(
                    Environment.getExternalStorageDirectory(), "HasilFoto");
                imagesFolder.mkdirs();
                Date d = new Date();
                CharSequence s =
                    DateFormat.format("MM-dd-yy hh-mm-ss", d.getTime());
                File image = new
                    File(imagesFolder, s.toString() + ".jpg");
                Uri uriSavedImage = Uri.fromFile(image);
                it.putExtra(MediaStore.EXTRA_OUTPUT, uriSavedImage);
                startActivityForResult(it, 0);
            }
        });
    }
}
```

Menyimpan hasil foto di directory yang bukan merupakan default dari android, seperti misalnya direktori

"HasilFoto".

Menyimpan dg memakai
putExtra

Memakai onActivityResult()

Mendeteksi aksi dari activity dapat memakai method onActivityResult() yang menangkap aksi dari startActivityForResult, sehingga diperlukan kode unik agar saat mengaktifkan intent, dan mendeteksi hasilnya dapat dikenali dengan memakai kode yang sama, hal ini dapat dipakai untuk melakukan operasi pada hasil capture foto, misalnya di tayangkan ulang di imageView program sebelumnya

```
private static final int kodekamera = 222;
```

1. meluncurkan kamera menggunakan suatu Intent khusus :

```
b1.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        Intent it = new
Intent(MediaStore.ACTION_IMAGE_CAPTURE);
        startActivityForResult(it, kodekamera);
    });
```

2. unique request code yang menandakan bahwa aplikasi yang meminta request camera.

```
@Override
protected void onActivityResult(int requestCode, int resultCode,
Intent data) {
    super.onActivityResult(requestCode, resultCode, data);
    if (resultCode == Activity.RESULT_OK) {
        switch (requestCode) {
            case (kodekamera):prosesKamera(data);break;
        }
    }
}

private void prosesKamera(Intent datanya){
    Bitmap bm;
    bm = (Bitmap) datanya.getExtras().get("data");
    iv.setImageBitmap(bm);
    Toast.makeText(this,"Data Telah Terload ke
ImageView",Toast.LENGTH_SHORT).show();
}
```

```
package dwi.anif.kamera;

public class MainActivity extends AppCompatActivity {
    Button b1;
    ImageView iv;
    private static final int kodekamera = 222;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1 = (Button) findViewById(R.id.button);
        iv = (ImageView) findViewById(R.id.imageView);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                startActivityForResult(it, kodekamera);
            }
        });

        @Override
        protected void onActivityResult(int requestCode, int resultCode, Intent data) {
            super.onActivityResult(requestCode, resultCode, data);
            if (resultCode == Activity.RESULT_OK) {
                switch (requestCode) {
                    case (kodekamera): prosesKamera(data); break;
                }
            }
        }

        private void prosesKamera(Intent datanya) {
            Bitmap bm;
            bm = (Bitmap) datanya.getExtras().get("data");
            iv.setImageBitmap(bm);
            Toast.makeText(this, "Data Telah Terload ke ImageView", Toast.LENGTH_SHORT).show();
        }
    }
}
```

Jalankan program foto ini akan sama dengan program sebelumnya, hanya saja, hasil capture kamera akan di tayangkan di imageview

Tidak memakai
putExtra

Menyimpan Foto ke File Lain

Dengan memakai program dibawah ini, sebenarnya hasil foto sudah terdapat di file gallery

```
Intent it = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
startActivityForResult(it,0);
```

Jika di inginkan hasil tersebut berada di folder atau file lain, maka diperlukan tempat yang harus di tentukan

```
private void prosesKamera(Intent datanya) throws IOException {
    Bitmap bm;
    bm = (Bitmap) datanya.getExtras().get("data");
    iv.setImageBitmap(bm);

    ByteArrayOutputStream stream = new ByteArrayOutputStream();

    bm.compress(Bitmap.CompressFormat.PNG, 100, stream);
    byte[] byteArray = stream.toByteArray(); // convert camera photo to byte array

    // save it in your external storage.
    File dir= Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_DCIM);
    File output=new File(dir, "simpan.png");
    FileOutputStream fo = new FileOutputStream(output);
    fo.write(byteArray);
    fo.flush();
    fo.close();
    Toast.makeText(this,"Data Telah Terload ke ImageView" + output,Toast.LENGTH_LONG).show();
}
```

Jangan Lupa Menambahkan permission menulis file

**Di AndroidManifest.xml
Atau check di setting ijin aplikasi**

**Tidak memakai
putExtra**

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
```

Akses Permission Write Storage

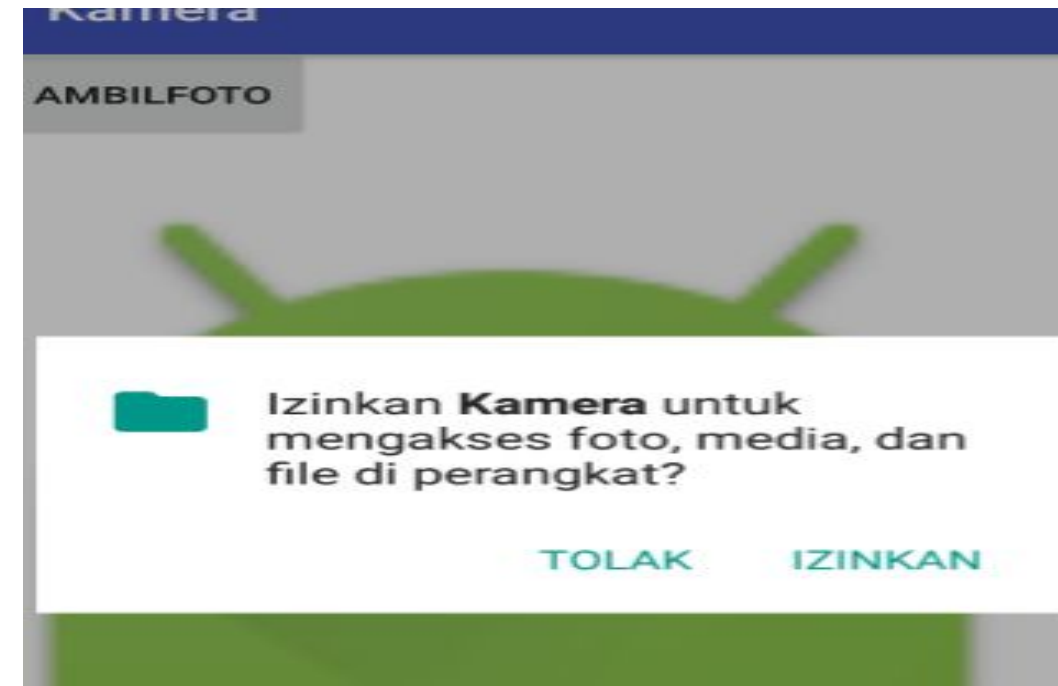
Requesting Permissions at Run Time

Beginning in Android 6.0 (API level 23), users grant permissions to apps while the app is running, not when they install the app. This approach streamlines the app install process, since the user does not need to grant permissions when they install or update the app. It also gives the user more control over the app's functionality; for example, a user could choose to give a camera app access to the camera but not to the device location.

```
public class MainActivity extends AppCompatActivity {  
    Button b1;  
    ImageView iv;  
    private static final int kodekamera = 222;  
    private static final int MY_PERMISSIONS_REQUEST_WRITE = 223;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        askWritePermission();  
        setContentView(R.layout.activity_main);  
    }  
}
```

```
private void askWritePermission() {
```

```
    if (android.os.Build.VERSION.SDK_INT >= 23) {  
        int cameraPermission = this.checkSelfPermission(Manifest.permission.WRITE_EXTERNAL_STORAGE);  
        if (cameraPermission != PackageManager.PERMISSION_GRANTED) {  
            this.requestPermissions(new String[]{Manifest.permission.WRITE_EXTERNAL_STORAGE},  
MY_PERMISSIONS_REQUEST_WRITE);  
        }  
    }  
}
```



Menyimpan Foto ke File Lain

Program sebelumnya memakai `intent.putExtra` untuk menyimpan hasil foto ke direktori dan nama file lain, dan dipanggil sebelum `startActivity`

```
Uri uriSavedImage = Uri.fromFile(image);
it.putExtra(MediaStore.EXTRA_OUTPUT, uriSavedImage);
startActivityForResult(it, kodekamera);
```

Pada method `onActivityResult`, tidak dapat memakai langsung hasil intent seperti sebelumnya

```
bm = (Bitmap) datanya.getExtras().get("data");
iv.setImageBitmap(bm);
```

Tidak ada data karena
sudah di `putExtra`,
sebelumnya

Program harus diubah untuk membaca file hasil foto berdasarkan path dan nama foto, baru kemudian di set ke `imageview`

`nmFile` adalah nama
File yg di deklarasi
global

```
private void prosesKamera(Intent datanya)
{
    Bitmap bm;
    //bm = (Bitmap) datanya.getExtras().get("data");
    BitmapFactory.Options options;
    options = new BitmapFactory.Options();
    options.inSampleSize = 2;
    bm = BitmapFactory.decodeFile(nmFile, options);
    iv.setImageBitmap(bm); // Set imageview to image that
    was
    Toast.makeText(this, "Data Telah Terload ke ImageView"
    +nmFile, Toast.LENGTH_SHORT).show();
}
```

Program Selengkapanya (1)

```
public class MainActivity extends AppCompatActivity {
    Button b1;
    ImageView iv;
    private static final int kodekamera = 222;
    String nmFile;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1 = (Button) findViewById(R.id.button);
        iv = (ImageView) findViewById(R.id.imageView);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent it = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                File imagesFolder = new
                File(Environment.getExternalStorageDirectory(), "HasilFoto");
                imagesFolder.mkdirs();
                Date d = new Date();
                CharSequence s = DateFormat.format("yyyyMMdd-hh-mm-ss", d.getTime());
                nmFile = imagesFolder + File.separator + s.toString() + ".jpg";
                File image = new File(nmFile);

                Uri uriSavedImage = Uri.fromFile(image);
                it.putExtra(MediaStore.EXTRA_OUTPUT, uriSavedImage);
                startActivityForResult(it, kodekamera);
            }
        });
    }
}
```

Program Selengkapnya (2)

```
@Override
protected void onActivityResult(int requestCode, int resultCode,
Intent data)
{
    super.onActivityResult(requestCode, resultCode, data);
    if (resultCode == Activity.RESULT_OK)
    {
        switch (requestCode) {
            case (kodekamera): prosesKamera(data); break;
        }
    }
}

private void prosesKamera(Intent datanya)
{
    Bitmap bm;
    //bm = (Bitmap) datanya.getExtras().get("data");
    BitmapFactory.Options options;
    options = new BitmapFactory.Options();
    options.inSampleSize = 2;
    bm = BitmapFactory.decodeFile(nmFile, options);
    iv.setImageBitmap(bm); // Set imageview to image that was
    Toast.makeText(this, "Data Telah Terload ke ImageView" + nmFile,
    Toast.LENGTH_SHORT).show();
}
```

Hasil bitmap di decode
selalu null, krn terlalu
besar shg karena harus
dikecilkan ukurannya
(skala)

Memakai API Camera

API Camera di sediakan oleh kelas Camera

Camera object = null;
object = Camera.open();

Fungsizx yg disediakan kelas Camera

No	Method & Description
1	getCameraInfo(int camerald, Camera.CameraInfo cameraInfo) It returns the information about a particular camera
2	getNumberOfCameras() It returns an integer number defining of cameras available on device
3	lock() It is used to lock the camera , so no other application can access it
4	release() It is used to release the lock on camera , so other applications can access it
5	open(int camerald) It is used to open particular camera when multiple cameras are supported
6	enableShutterSound(boolean enabled) It is used to enable/disable default shutter sound of image capture

Android Working with Camera2 API

Compared with old Camera API, new Camera2(`android.hardware.camera2`) is much complex as it involves all asynchronous calls. Android API 21 introduced this new Camera2 API. So, if you are working on API less than 21, use the old Camera API(`android.hardware.camera`) and for all above API 21 use this API as the old one is deprecated. Camera2 works in the following way:-

- **CameraManager**- get all camera's available in the device each having cameraId like front camera and back camera and its properties can be obtained from CameraCharacterstics class. (Select Camera and Create CameraDevice)
- **CameraDevice**- get camera device from cameraManager class by its id and perform related operations. (Create CaptureRequest and Create CameraCaptureSession)
- **CaptureRequest**- Create a CaptureRequest from cameradevice to capture images. (Link Surface for Viewing and Make CaptureRequest)
- **CameraCaptureSession**- Create a CameraCaptureSession to get CaptureRequest's from Camera Device. (Capture Camare Image and put the result on the Surface registered in CaptureRequest.)
- **CameraCaptureSession.CaptureCallback**- It will provide all the capture session results.

One of reasons why the related objects are a little bit complex is most of set-up process is done asynchronously. For example when we create CameraDevice, at first we register a callback function on CameraManager. When CameraDevice is created and the callback is called, we get CameraDevice in the callback code.



Terima Kasih

